

Inventory Tracking System for Non-Profits

A free, open-source web platform to help food banks track inventory, forecast demand, and reduce waste

React + FastAPI + PostgreSQL ML Demand Forecasting Barcode Scanning Open Source - MIT licence

THE PROBLEM

Non-Profits Are Stuck With Inefficient or Expensive Tools

Food banks and other non-profits often manage hundreds of donations and inventory items using pen-and-paper or basic spreadsheets. This leads to:

- ✗ Items expiring before they can be distributed
- ✗ No visibility into what supplies will be needed next week



At the same time, existing software solutions introduce new challenges:

- ✗ Expensive paid software that cut into charitable budgets
- ✗ Volunteers requiring significant training on complex systems

OUR OBJECTIVE

Build a Free, Easy-To-Use System Anyone Can Deploy



We set out to replace costly and clunky tools with a modern web app that any non-profit can use for free. Our core goals:

Track inventory

Scan barcodes, log items, and see what's in stock in real time

Data Insights

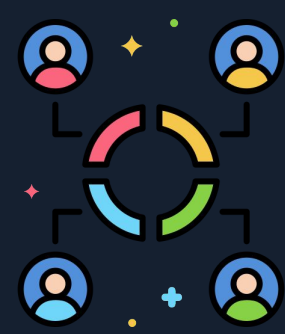
Use charts and ML to forecast what items will be needed soon

Stay free

Open-source under MIT licence - no ongoing costs for the organizations

OUR STAKEHOLDERS

Who Does This System Serve?



Volunteers & Staff

Primary users - scan items, update inventory, check alerts. Needs zero training to get started.

Non-Profit Management

Use dashboards and forecasts to plan restocking and allocate resources effectively.

Donors

Benefit from transparent tracking that shows their contributions are used well.

Community Members

End recipients who benefit from faster distribution and reduced waste.

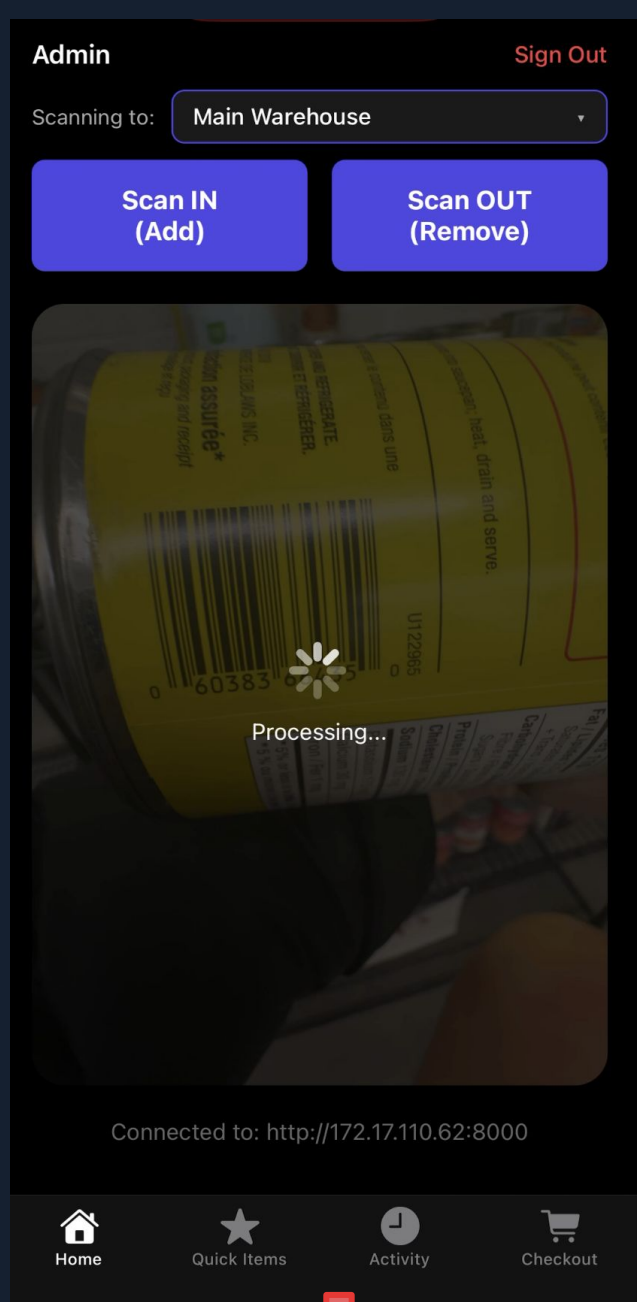
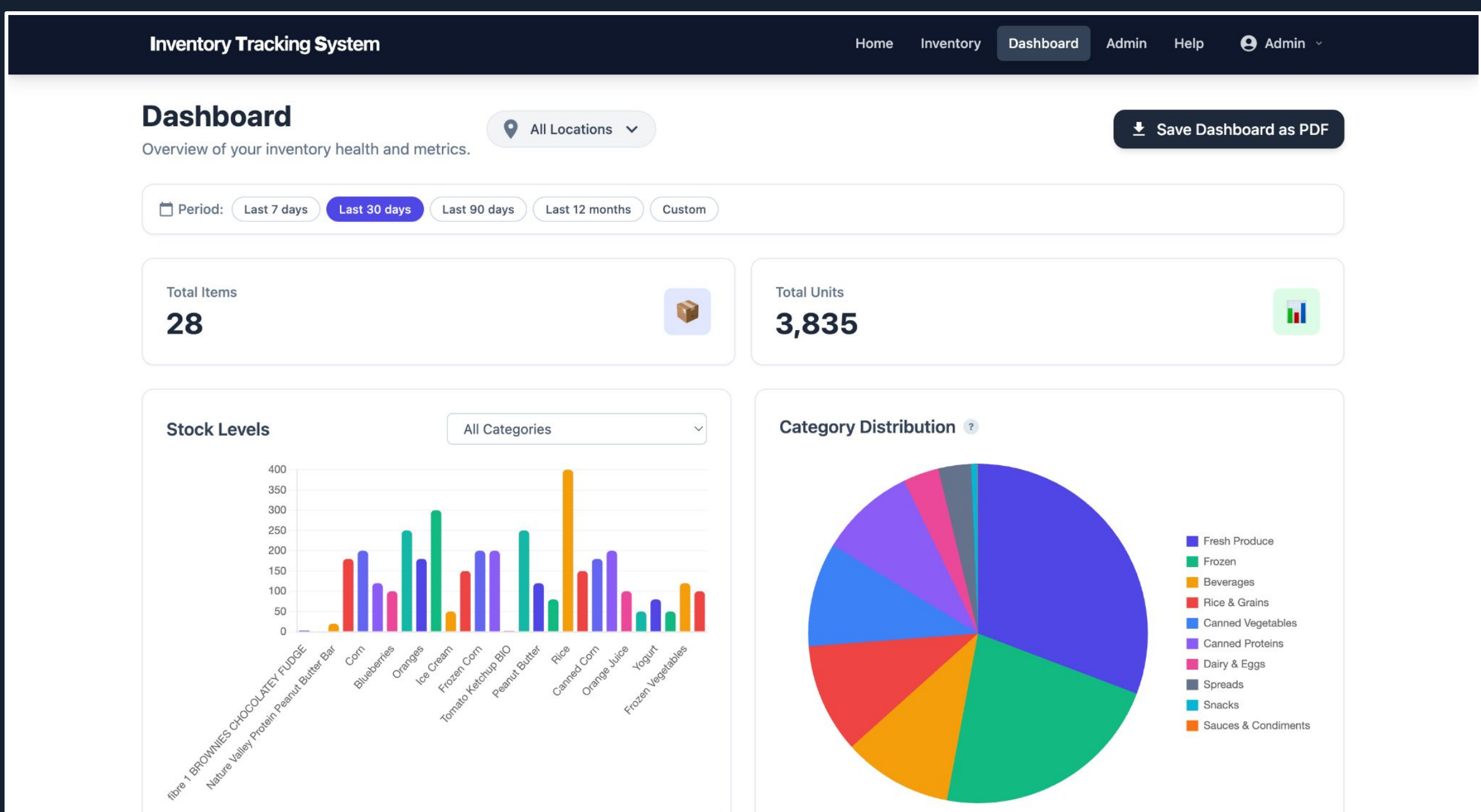
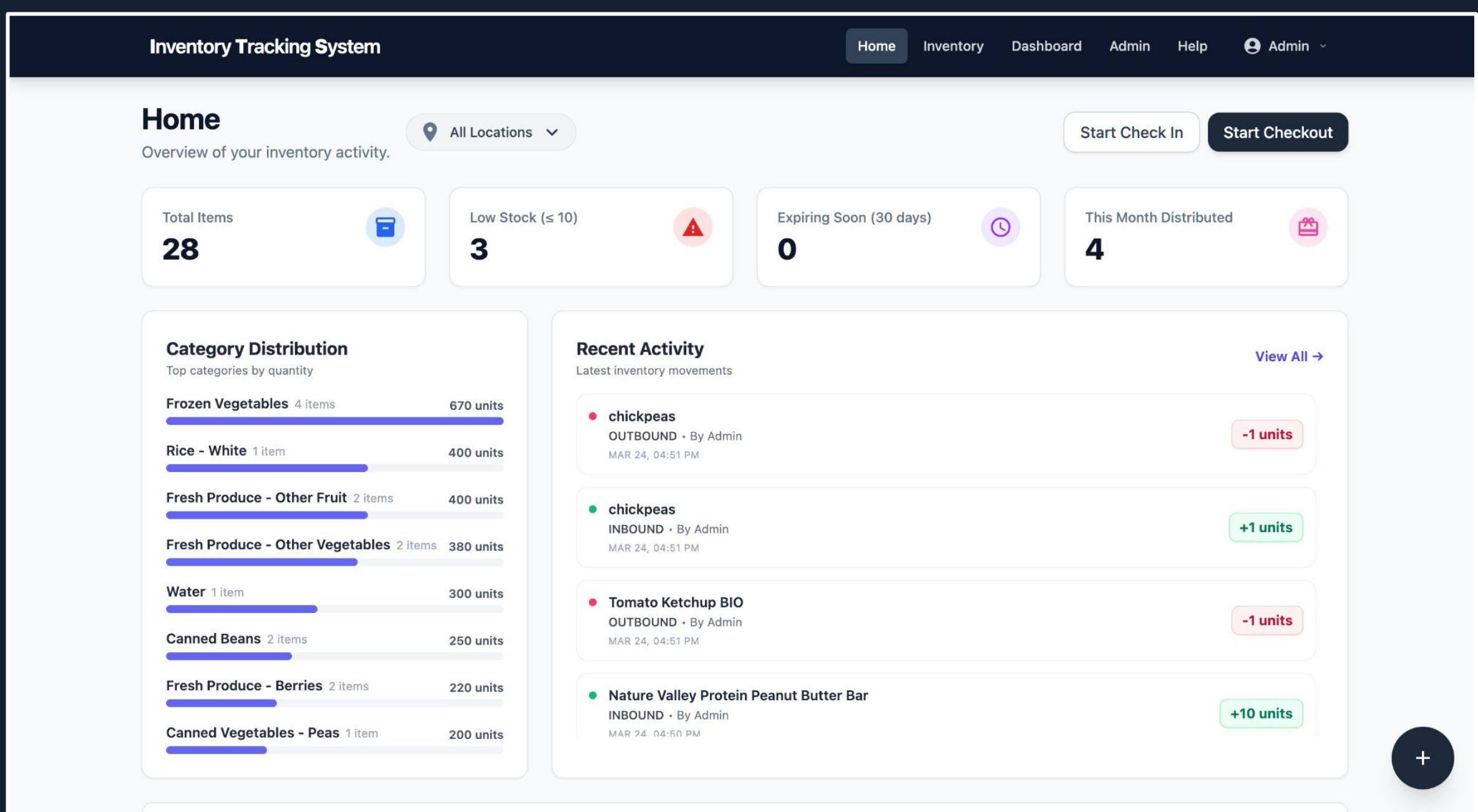
CONCLUSION

We built a **free, open-source inventory system** that gives non-profits the tools previously only available through expensive paid software. With **barcode scanning, inventory workflows, ML-driven demand forecasting**, and an **intuitive dashboard**, food banks can cut down on waste, plan smarter, and spend more energy on the communities they serve.

RESULTS

What We Built

We delivered a fully working web application with a mobile companion app.



✓ Full inventory management with **barcode scanning** and **bulk CSV upload**

✓ Customizable dashboard with **low-stock alerts**, **expiry warnings**, and **PDF export**

✓ Role-based login (**volunteer, staff, admin**) with JWT authentication

✓ **Mobile app companion** for rapid barcode scan-in and scan-out

✓ **Demand forecasting pipeline** to predict output

TECH STACK & DEMAND FORECASTING PIPELINE

Modern stack, lightweight by design

React + Vite Python FastAPI PostgreSQL JWT Auth Pandas / NumPy

- 1 Historical data** — Inventory movement data is stored and used to generate insight
- 2 Weekly demand** — Outbound inventory is grouped by category into continuous weekly demand history
- 3 Model applied** — An exponential smoothing model is trained to capture trends in weekly output
- 4 Predictions served** — Forecasts shown on dashboard, along with other customizable charts about inventory needs